



DWAN: dynamic weighted adversarial network for medical image segmentation and classification

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Abstract

Supervised classification of image segments is a key component of several healthcare image segmentation techniques. When the training and test images are comparable, these techniques typically perform well. However, when the training and test images differ, which might happen because of differences in scanners, scanning techniques, or patient groups, issues may arise. Results can be improved in these situations by providing training images that resemble the test images. However, this method does not make dissimilar data more similar; it only works if some training images have previously been comparable to the test data. We investigate ways to enhance image weighting by minimizing the discrepancies between training and test images using a method known as kernel learning. Additionally, we present Dynamic Weighted Adversarial Network (DWAN), a novel image identification technique that reduces the disparity between training and test data. This DWAN technique enables simultaneous optimization of the image kernel and weight. Experiments on several brain sample image kinds demonstrate that dynamic weighting by itself greatly enhances performance on a range of data. The performance of the novel DWAN weighting technique is comparable to that of the current techniques. Whether optimized independently or together, image weighting and dynamic models can offer a slight performance improvement.

Keywords Deep learning (DL) · Prediction · Weighted model · Brain image and optimization

1 Introduction

Medical research and clinical treatment both depend on the classification of biological samples into different tissues and patterns. Automated classification is significant because directly categorizing 3-D images takes a lot of time and is subject to intra- and inter-observer fluctuation. Voxel-wise supervised classification is the foundation of many automated segmentation techniques (Thukral 2015). Here, a classification algorithm is trained using subjectively interpreted training data that identifies which class (which tissue or structure) each voxel belongs to. For instance, entire brain identification, nervous tissue recognition, white matter lesion identification, and structure of the brain segmentation have all used such techniques in brain imaging (MRI) (Lee et al. 2017). The use of multiple scanners, imaging procedures, or patient categories can all cause performance to deteriorate when training and test data differ. This is a drawback of supervised learning techniques. Human observers can readily adjust to these variations. However, because supervised learning techniques frequently produce different organization of training and test instances in the feature region, they may have trouble with image discrepancies (Huang and Frangi 2017). Techniques for transfer learning are made to deal with variations in sample distributions and other aspects of training and test data. Weighting training samples is the foundation of several transfer learning techniques that have been introduced for healthcare image classification. Weighting individual images, or entire training samples can do this. The benefit of image weighting over sample weighting is that it can tolerate scanner variations without the need for labeled training image samples from the test optical scanner. No efforts are made to lessen the disparities in the feature representation (i.e. the feature counts) of training and test images; these previously proposed weighting techniques solely weight training samples in their current state. A training image can only be used as

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is (given a positive weight) or not at all (given a weight of zero) in an image weighting method (Sevetlidis and Giuffrida, and S. A. Tsafaris 2016). (1) Make the feature distributions between training and test samples more similar and (2) as effectively separate the various classes as possible, we thus suggest combining image weighting using a feature representation transfer phase (Joyce and Tsafaris 2017).

Explicit feature representation transfer to healthcare image segmentation has not been extensively studied up to this point. However, several techniques have been developed in the fields of computer vision and machine learning. A lot of them look for a linear adjustment that reduces the disparity in organization between test and training data (Chartsias et al. 2018). Transformations are frequently carried out in a high- (potentially infinitely-) dimensional kernel space since linear transformations are frequently insufficient for addressing dataset discrepancies (see, for instance, the metric learning techniques). As an alternative, a kernel that lessens distribution disparities could be learned directly. For instance, the author in Jog et al. (2017) suggests an unsupervised methodology for learning a kernel that reduces the maximum mean discrepancy (MMD) among training and test images, hence increasing the similarity of training and test distributions. However, the trained kernel is not optimum for segmentation because the approach is unsupervised. A supervised approach is presented by the author in Zhao et al. (2017) that learns a kernel, which minimizes inaccurate segmentation on a few labeled sample data in the learned kernel space while also minimizing MMD between training data and test images. Unfortunately, training on a large number of samples (greater than one thousand) is computationally challenging for these 2 approaches, as well as the majority of kernel learning and metric learning techniques that are in use nowadays (van den Berg, and I. Isgum 2017). However, there are numerous additional samples for training available for voxel-wise categorizing, and employing them will probably yield superior results. For instance, the image weighting model presented in Ben-Cohen et al. (2017) employs fifty thousand samples for a single image.

In this work, researchers provide effective kernel learning techniques that were applied in image weighting. We start by talking about a multiple kernel learning (MKL) technique that improves distances among classes while minimizing distances within them. Second, we apply our MKL technique with an MMD term to the structure given in Mahmood and Durr (2018). In addition to producing a kernel space that is appropriate for categorizing, this also improves the similarity between the training and test sample distributions. The suggested techniques may be learned using 10,000 training samples extracted from a single image sample (Huang and Wang 2018), in contrast to the approaches. We examine if employing these kernel learning techniques

improves segmentation over utilizing a conventional Gaussian kernel (Hiasa et al. 2018). We also look into whether applying image weighting enhances performance with these learned kernels. We also demonstrate that the image weights and kernel may be found using the MMD metric (Costa et al. 2018; Calimeri et al. 2017). This makes it easier to optimize by allowing the kernel and image weights to be improved together.

This research outlines image weighting techniques: an innovative image weighting technique that decreases MMD and two weighting techniques that reduce the divergence. The two MKL approaches that were employed: centered kernel alignment and the MMD multiple kernel learning architecture are covered. We also demonstrate the integration of MMD image weighting and Dynamic Weighted Adversarial Network (DWAN) scheme. Using data from several datasets that were obtained using various scanners and scanning techniques, we examined the effectiveness of MKL, image weighting, and the integration of two techniques during the training and testing process. White matter lesion classification, hippocampal categorization and brain tissue classification were the medical image classification tasks were examined.

The work is drafted as: Sect. 2 depicts the comprehensive analysis of diverse approaches. The methodology is explained in Sect. 3. The experimental results are shown in Sect. 4 and the conclusion is depicted in Sect. 5.

2 Related works

This section of the study will look in greater detail at how different ML approaches are used to analyze brain tumors. The several types of ML approaches and algorithms that can be used for brain tumor classification are described in Huo et al. (2018). We emphasize how important method improvement is to the advancement of diagnostic imaging and how it leads to improvements in diagnostic abilities and the use of treatments. Aspects of collecting and choosing features are also examined, in addition to various machine learning algorithms that aid in accurately classifying brain tumors from imaging data. We intend to determine the advantages, disadvantages, opportunities, and threats of this thorough examination of using machine learning in the classification of brain tumors (Zhao et al. 2018). The research “A Multiclass Tumor Assessment of High-Grade Malignant Brain Tumors” uses a computer-assisted diagnosis approach to categorize high-grade invasive brain tumors into different classes (Mardani et al. 2019). To optimize the quantity of data that can be extracted from the results, the suggested feature selection technique uses a very broad feature from six areas and allows the differentiation between five distinct

types of malignant brain tumors. To make it easier to choose appropriate features from the feature set pool, a novel method called CVM is used for feature selection (Quan and Jeong 2018; Yu et al. 2018).

Consequently, three classifiers are trained and assessed in terms of multiclass classification accuracy using the previously indicated selected features. The experiments' outcomes are represented as having excellent performance. In contrast to existing algorithms like ICA and GA with NN, KNN, and mSVM (Wang et al. 2018), the suggested approach remarkably shows an enhancement in precision of roughly 2 percent, 3 percent, and 4 percent, respectively. In particular, it is feasible to note that the NN classifier exhibits the maximum effectiveness in the tests performed (Nie et al. 2018), with an accuracy rate of 95%. The following diagram demonstrates the effectiveness and proficiency of the suggested methodology in grouping MBT into several categories (Mao et al. 2017). ML approach that is hybridized utilizing data and a firefly method based on MAP, texture, intensity, and shape features are extracted. With three distinct locations identified throughout the tumor identification process and MRI samples of brain carcinoma and strokes divided into 4 groups, the proposed method achieves a reliable accuracy of 99.2% for brain strokes and 88.3% for brain tumors (Murugan 2017). The hybridized support vector-based random forest classifier (HSVFC) performs better, achieving an excellent F-score of 0.91 and a minimal False Positive Rate (FPR) of 0.06 for classifying brain tumors and an F-score of 0.99 and 0.0 FPR for classifying brain strokes (Hinz et al. 2018). Furthermore, HSVFC outperforms DC, KNN, SVM, ResNet-18, and FFNN classification algorithms; it can classify brain tumors with an overall mean accuracy of 76.55% and brain strokes with the highest mean accuracy of 98.17%. Utilizing its multicolored tissue representation, the investigation aims to understand the issue. Clustering procedures are optimized by the use of the firefly method (Kim and J. P. Haldar 2018). The study highlights the need for thorough information about the hyper-spectral imaging dataset, a comparison with other tumor localization techniques, an examination of multiple datasets, and a review of the outcomes' interpretability to bridge the research gap (Yang et al. 2018).

The outcomes depict that the developed approach is efficient, surpassing earlier methods in tumor localization precision with 96.48 percent accuracy, 96.33 percent sensitivity, and 98.25 percent specificity. BTc Based on magnetic resonance imaging use previously trained DNNs. Following evaluation by numerous machine learning classification algorithms, the top 3 deep characteristics are combined to create an ensemble. The study uses three widely accessible brain MRI records for evaluation, but it also points out several research gaps, including the need for the suggested

ensemble of advanced characteristics to be interpretable and explicable, the lack of comparison with DL algorithms designed for medical image processing, and the lack of comprehensive dataset. Although the experimental results indicate encouraging performance gains, the lack of exact measures and comparisons with other advanced methodologies makes it impossible to completely assess the results' applicability and effectiveness (Jiang et al. 2018).

The use of machine learning models for brain tumor identification is examined in the publication. The authors provide their experimental results and address prior studies that used models based on ML trained on domain-driven datasets for tumor recognition, emphasizing the characteristic of brain tumors as a group of cells with unlimited progression. The article does not specify which specific research gap is being addressed, even though it builds on previous work in recognizing brain tumors using ML algorithms. However, by using ResNet 5.0 with a Transfer Learning Model, the authors can identify Glioma, Meningioma, and pituitary tumors with 96% accuracy (Turner et al. 2018). They observe that this algorithm's efficacy in classifying brain tumors was demonstrated by the fact that it outperformed six distinct machine-learning approaches that were tested on their dataset. A research investigation was carried out that used two-dimensional magnetic resonance (MR) images to diagnose brain tumors earlier. The work does not explicitly define the research gap that demands to be addressed, despite using DL-based segmentation approaches on two-dimensional magnetic resonance (MR) images to address the requirement for early brain tumor identification. The actual challenge, however, is determining which combination of classifiers and deep learning feature extractors would increase the effectiveness of brain tumor detection systems (Gözcü et al. 2018). The results show that the top-performing model has a tenfold cross-validation accuracy of 99.39%. Studies emphasize designing augmentations that respect organ geometry or clinical appearance (e.g., domain-guided augmentation frameworks), since unrealistic transforms can harm model calibration. Recent work provides tools and guidelines to craft medically plausible augmentations for improved generalization. Transfer learning for medical imaging broadly means re-using weights or representations learned on a source task (usually ImageNet or large unlabeled medical volumes) and adapting them to a medical target task via fine-tuning or feature transfer. Two recurring themes in the literature are (a) whether ImageNet pretraining helps for medical data, and (b) whether self-supervised / medical-domain pretraining (3D where relevant) is superior.

3 Methodology

Image weighting is carried out by assigning a weight to every training image based on reducing the measurement of distance among the test image for segmentation and the training images' probability density functions (PDFs). Then, the weight allocation of all training voxels and their related class labels are sampled to create a training dataset relative to the training samples and their assigned weights. Therefore, test images are segregated by voxel-wise segmentation and a classification algorithm is trained. Each test image undergoes separate kernel learning and the learned kernel is then used for segmentation using a kernel model. Some characters (like x) to indicate column vectors and matrices and (like y, θ) to indicate scalar values. Using the kernel trick, a distinction is produced for φ which indicates a mapping into the region of the kernel and K which indicates a kernel ($K(x, y) = \varphi(x)^T \varphi(y)$). Subscripts are used to indicate vectors and matrices such as x_i, M_p and q . For example, n^{te} indicates the total amount of verification data and n^{tr} indicates the amount of training data in a single image. The superscripts are additionally employed for identification. There are m training images utilized in total, each of which may have unique scanning properties. Some randomly chosen training samples (voxels) n^{tr} are provided by each training image $x_j^{m_i}$ where (j takes the values from 1 to n^{tr}), where $x_j^{m_i} \in R^n$ indicates a vector with a value that represents each of the n attributes. Here, X^{tr} represents the $m \cdot n^{tr} \times n$ matrix of all training samples. Here, $n^{te} \times n$ test voxels x_j^{te} , ($j = 1, 2, \dots, n^{te}$) are obtained from the test sample, for which we identify the label y_j^{te} . Each sample $x_j^{m_i}$ has a label $y_j^{m_i}$ is the element of N . X^{te} is used to represent the test voxels with $n^{te} \times n$ matrix. The distribution $P^{te}(x)$ is followed by the test data and $P_{m_i}(x)$ by the training data from image sample m_i . The above distributions are used to determine the weight of m_i which yields $w = [w_1, w_2, \dots, w_m]^T$, the weight vector for every training sample. Since these weights are standardized and non-negative in nature $\sum_{m_i} w_{m_i} = |w| = 1$.

3.1 Dynamic weight analysis

A Dynamic Weighted Adversarial Network is justified for brain image segmentation and classification because it directly addresses tumor heterogeneity, class imbalance, noise/artifacts, joint multi-task demands, and patient-specific variability, all of which are dominant challenges in brain imaging but less critical in general medical image segmentation tasks. The adversarial component of DWAN is the Weight-based domain discriminator loss that minimizes

the distributional gap between source and target features by training a critic to estimate the Wasserstein distance and a feature extractor to adversarially reduce it. A whole training PDF is computed as a weighted sum of all the independent training PDFs produced by assigning a weight w_i to each training image m_i :

$$P^{tr}(x) = \sum_{m_i}^m w_{m_i} P_{m_i}(x) \quad (1)$$

By reducing a convex space metric between the entire training PDF and the verification PDF, the ideal weights w^* are determined:

$$w^* = \arg \min_w dist_w(P^{tr}, P^{te}) \quad (2)$$

The research describes an approach that uses the estimation of kernel density to directly estimate the PDFs P_{m_i} (based on the distance function, possibly P^{te}) before applying a distance function to the calculated PDFs. Using a Gaussian kernel and kernel density estimation, these PDFs are estimated from the data. Silverman's formula is used to find the kernel parameter, σ^S .

$$\sigma^S = \left(\frac{4}{n+2} \right)^{\frac{1}{n+4}} \cdot n^{tr \frac{-0}{n+4}} \sigma^{tr} \quad (3)$$

In this case, σ^{tr} is the approximate standard deviation of the training images, averaged across all the extracted features. For a multiple-variate Gaussian kernel, the selected σ^S reduces the mean integral square error that exists between the predicted and real PDF. The distance between training and test PDFs can be measured in several ways. One of these is the KL divergence, which is described in Eqs. (4)–(5):

$$KL(P^{te} || P^{tr}) = \int_D P^{te}(x) \log \left(\frac{P^{te}(x)}{P^{tr}(x)} \right) dx \quad (4)$$

$$\approx \frac{1}{n^{te}} \sum_{j=1}^{n^{te}} \log P^{te}(x_j^{te}) \quad (5)$$

$$\approx \frac{1}{n^{te}} \sum_{j=1}^{n^{te}} \log \left(\sum_{m_i=1}^m w_{m_i} P_{m_i}(x_j^{te}) \right) \quad (6)$$

where, D is P^{tr} and P^{te} domain. Recall that Eq. (6) first term is independent of w . Therefore, increasing $\sum_{j=1}^{n^{te}} \log \left(\sum_{m_i=1}^m w_{m_i} P_{m_i}(x_j^{te}) \right)$ with the conditions $w \geq 0$ and $|w| = 1$ yields the ideal weights w . The standard

deviation (σ^S) is sensitive to outliers. Taking the minimum ensures that the bandwidth is not overinflated in the presence of outliers, leading to a smoother but still representative kernel density estimate. So, σ^S is essentially a robust measure of spread, combining standard deviation and interquartile range. KL divergence is an asymmetric information-theoretic measure of divergence. By symmetrizing KL (via geometric averaging) and focusing on overlap rather than log-ratios, one arrives at the Bhattacharyya coefficient and its logarithmic form, the Bhattacharyya distance, which is symmetric and bounded. The Bhattacharyya distance (BD) is a second technique for calculating the separations between PDFs that did well in Eq. (7):

$$BD(P^{te}, P^{tr}) = - \int_D \sqrt{P^{te}(x) P^{tr}(x)} dx \quad (7)$$

$$\approx \frac{1}{n^{tr}} \sum_{x_i \in D} \sqrt{P^{te}(x_i)} \quad (8)$$

where, D is used to randomly select the x_i . Equation (8) criterion is reduced for w under the condition that w is greater than or equal to 0. and $|w|$ equals 1. A novel image weighting technique based on maximal mean discrepancy (MMD) reduction is suggested. This approach eliminates the need for a PDF estimation stage since it can instantly ascertain the image weights from the examples provided. The MMD image weighting assigns weights to groupings of samples (images) rather than individual samples. The MMD in all training voxels (X^{tr}) and test voxels (X^{te}) is minimized to get the image weights (w). The space between the mean of x and y samples in a kernel-based space φ ,

$$MMD(X, Y) = \left\| \frac{1}{n^X} \sum_{i=1}^{n^X} \varphi(x_i) - \frac{1}{n^Y} \sum_{j=1}^{n^Y} \varphi(y_j) \right\|^2 \quad (9)$$

where, n^X and n^Y . represent the quantity of samples x and y , respectively, is known as the MMD between the datasets X and Y . By reducing the MMD between the test set X^{te} and the training dataset X^{tr} which is assigned as a single image weight, the image weights are established:

$$MMD(X^{tr}, X^{te}) = \left\| \frac{1}{n^{tr}} \sum_{m_i=1}^m w_{m_i} \sum_{j=1}^{n^{tr}} \varphi(x_j^{m_i}) - \frac{1}{n^{te}} \sum_{i=1}^{n^{te}} \varphi(x_i^{te}) \right\|^2 \quad (10)$$

$$= \frac{1}{n^{tr2}} w^T K^{tr, tr} w - \frac{2}{m^{tr} n^{te}} w^T k^{tr, te} + \frac{1}{n^{te2}} k^{te, te} \quad (11)$$

In this case, vector $K^{tr, tr}$ contains the length m of internal products between each training image and the test sample, $K^{tr, te}$ is an integer providing the internal product of the test

samples and $K^{tr, tr}$ is a $m \times m$ matrix of internal products between training images. The additional resources, Eq. (11), include the procedure for $K^{tr, tr}$, $K^{tr, te}$ and $K^{te, te}$ a (further resources that are accessible in the files). Equation (12) can be expressed as:

$$MMD(X^{tr}, X^{te}) = \omega^T M \omega \quad (12)$$

where, ω^T is declared as $[\omega^T, 1]$ and under the restrictions $w \geq 0$, $|w| = 1$, the ideal image weights can be determined by reducing Eq. (12) for ω provided a kernel K . Our goal is to recognize a kernel space that reduces dataset discrepancies. For the matrix we find to be a kernel (according to Mercer's axiom), we would have to impose requirements to ensure that it is symmetric and positive semi-definite. MKL begin with a set of n^k pre-determined fundamental kernels K_k^b (where $k = 1, 2, 3, \dots, n^k$), is a reasonably simple method for locating such a kernel matrix. The best linear combination of these basis kernels is identified as the searched kernel:

$$K = \sum_{k=1}^{n^k} v_k K_k^b \quad (13)$$

where, v takes the value from $[v_1, v_2, v_3, \dots, v_{n^k}]^{T \geq 0}$ is the kernel weight. Here, K must also be a kernel since it is a linear arrangement of kernels. This work introduces kernel learning techniques. The MMD-KL technique is comparable to the technique proposed and the centered kernel alignment. To jointly maximize the kernel and image weights, this technique can be effectively used with MMD image weighting. For training samples to appear similar to the samples of the test dataset in the learned kernel address space, the investigators suggest finding K that reduces (1) the gap existing in the distribution of training and a test dataset, and (2) a certain amount of segmentation faults on the training images, which makes the learned kernel is appropriate for segmentation. The following optimization criterion is the outcome of this:

$$K^* = \arg \min_K \theta dist_K(P^{tr}, P^{te}) + f_K(X^{tr}, y^{tr}) \quad (14)$$

where, f_K quantifies an error in classification in the K (kernel space), $dist_K$ represents a PDF distance function in the K 's kernel memory and θ is a parameter that trades off both terms. Some researchers employ either the Hinge loss or the structural risk functional for f_K , both of which are computationally quite costly if a large number of training samples are employed. Figure 1 shows that, DWAN model.

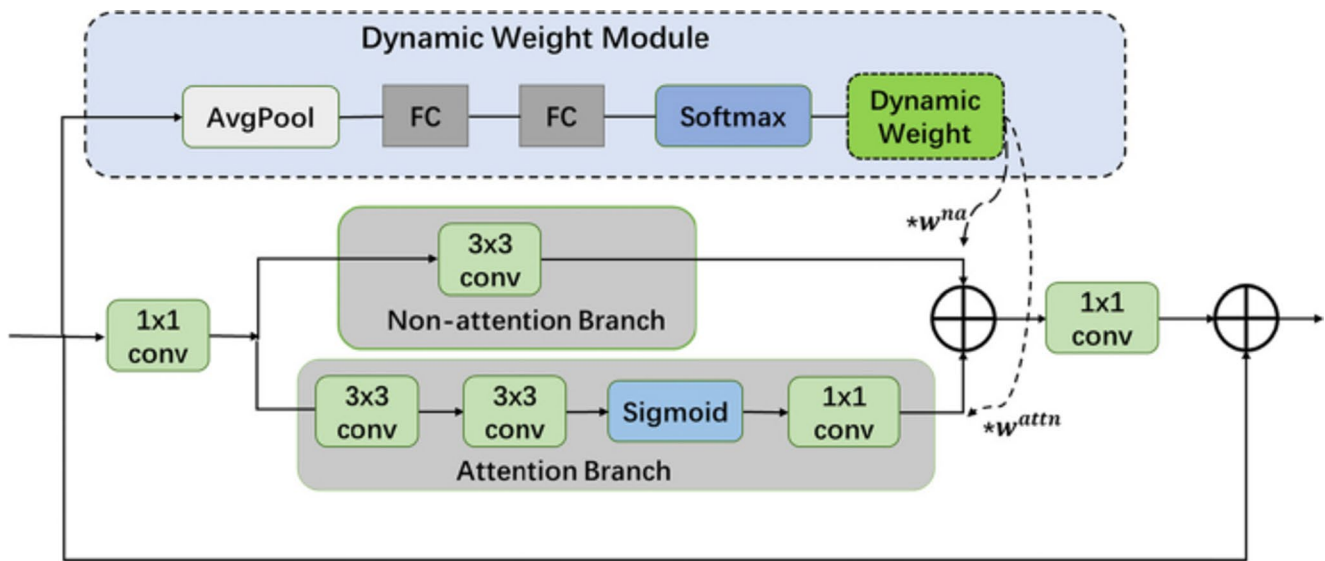


Fig. 1 DWAN model

3.2 Adversarial learning

A neural network designs known as generative adversarial networks (GAN) are made up of two sub-networks: a discriminator and a generator. In a target domain, it learns to distinguish between the produced image G and the actual image D by mapping a latent variable z (usually random noise) to an image y . Both G and D are concurrently taught during GAN training, with G trying to produce images that are identical to the real ones and D trying to distinguish between generated and genuine images. If we only have T1 scans but need T2 scans for multiple sclerosis lesion detection $\rightarrow$ multi-contrast generation problem. Multi-contrast generation solves the missing modality problem (image completion). Segmentation domain adaptation solves the distribution shift problem (generalization). This can be accomplished by using the adversarial loss function (L_{GAN}) as follows:

$$L_{GAN}(G, D) = E_y [\log D] + E_z [\log (1 - D(G(z)))] \quad (15)$$

The anticipated value is indicated by E . G aims to enhance modeling high-spatial-frequency information by minimizing and maximizing the adversarial loss. At the same time, both G and D are trained concurrently. After convergence, it may generate realistic-looking fake images that G are unable to identify. A squared loss can be used in place of the adverse log-likelihood price for the adversarial loss in Eq. (15) to improve the stability of the training procedure. The generated network is trained using the following adversarial loss function:

$$L_{GAN}(G, D) = -E_y [(D(y) - 1)^2] - E_z [D(G(z))^2] \quad (16)$$

$$L_{cond-GAN}(G, D) = -E_{x,y} [(D(x, y) - 1)^2] - E_{x,z} [D(x, G(x, z))^2] \quad (17)$$

where, the source image is represented as x . When similar anatomy is collected under several distinct sample contrasts (such as T1- and T2-weighted images), MR imaging presents an equivalent challenge to the image-to-image translation process in automated vision. Here, we used conditional generative adversarial networks to create magnetic resonance images of an intended contrast given as a source, motivated by the recent achievements of adversarial networks. We looked at two various approaches for multi-contrast MR image generation to provide a complete solution. Initially, we assumed that the source and target contrast images were fully registered. Motivated by the pix2pix construction, we suggest a plan for this kind of scenario, which adds a pixel-wise loss to the objective function.

$$L_{L1}(G) = E_{x,y,z} [\|y - G(x, z)\|_1] \quad (18)$$

where, L_{L1} represents the pixel-wise loss function (L1). The latent variable was eliminated from the framework since it was found that the generator G in GAN ignored it. By focusing the networks not just on y but also on z the nearby cross-sections of x , we were able to generate each cross-section x by utilizing correlated data from adjacent cross-sections. The following is obtained by combining the adjacent cross-sections Eq. (19)–(21):

$$L_{cond-GAN-k} = -E_{x_k,y} [(D(x_k, y) - 1)^2] - E_{x_k} [D(x_k, G(x_k))^2] \quad (19)$$

$$L_{L1-k}(G) = E_{x_k, y} [||y - G(x_k, z)||_l] \quad (20)$$

$$L_{perc-k}(G) = E_{x_k, y} [||V(y) - V(G(x_k))||_l] \quad (21)$$

where, $x_k = [x_{-\frac{k}{2}}, \dots, x_{-2}, x_{-1}, x, x_{+1}, x_{+2}, \dots, x_{+\frac{k}{2}}]$ is a vector consisting of k successive cross-sections ranging from $-\frac{k}{2}$ to $\frac{k}{2}$; with the cross section x in the central point, and $L_{cond-GAN-k}$ and L_{L1-k} are the equivalent pixel-wise and adversarial loss functions. The aggregate loss function is represented as:

$$L_{L1}(G) = E_{x, y, z} [||y - G(x, z)||_l] \quad (22)$$

where, L_{GAN} is the entire loss function, λ manages the comparative weighing of the pixel-wise loss, and λ_{perc} manages the comparative weighing of the perceptual loss.

Input: samples from the provided input dataset;

Output: samples of fake or generated images

1. **For** all iter num **do**
2. **For** all count_num of samples **do**
3. Generation of fake images
4. Generation of real images from the dataset
5. Train the dataset with weighted parameters
6. **End for**
7. Generate fake images based on generator input;
8. Train the updated weight;
9. **End for**

Algorithm 1

4 Numerical results and discussion

Three MRI brain identification processes including white matter lesion (WML) categorization, classification of brain tissue, and hippocampus recognition were used in our voxel wise categorization tests. Each voxel within a subjectively labeled brain mask was categorized as white matter (WM), cerebrospinal fluid (CSF), and gray matter (GM) to categorize the brain tissue. Every voxel within an autonomously

produced brain mask was categorized as WML or non-WML for WML differentiation. Using multiple-atlas registration, ROI surrounding the hippocampus was identified for hippocampus differentiation. Each voxel within this ROI was categorized as either hippocampal or non-hippocampal. DWAN classifier based on intensity features was used for categorization. In an additional investigation, we evaluated DWAN efficiency. We gathered data from various datasets that were obtained using various scanners and scanning parameters for each of the three applications. In leave-one-out cross-validation, each sample was divided into segments once. The training data included entire dataset images except the test sample. The features employed, data, and experimental design are described in this section.

4.1 Dataset description

Table 1 offers a summary of all the data involved in the studies. The research utilizes 61 images obtained from 5 source data samples with comparable labor-intensive categorizations for brain tissue (BT) classification. Two datasets were from the Internet Brain Segmentation Repository (IBSR), one from the MRBrainS Challenge, and two from the Rotterdam Scan Study (RSS). While BT4 and BT5 were obtained using numerous scanners (although it is unclear which images are from the identical scanner), the images from the BT3, BT2, and BT1 datasets were obtained using a specific scanner. The brain tissue and mask have been manually segmented in every image. The cerebellum is part of the brain mask images present in BT3 and BT4 datasets, but not in images BT1, BT5, or BT2. In contrast to the T1 image samples from the previous experiments, the Haste-Odd images from BT2 exhibit reversed intensities. Consequently, before calculating the features, the voxel values in the BT2 images were reversed. A total of 40 manually segmented images from three datasets were employed for WML classification. MS Lesion Classification Challenge of MICCAI is used as a source dataset to gather two datasets and one from the RSS. The brain masks were created using brain extraction technology (BET). Due to their visual similarity, we considered the PD samples of WML1 as similar modality as the WML2 and WML3 T2 images for feature computation. We employed magnetic resonance image samples extracted from the Harmonized Protocol (HarP) for the hippocampal tests. The Alzheimer's disease Neuroimaging Initiative (ADNI) samples comprise 135 T1-weighted images with carefully labeled hippocampus. A total of 46 separate scanners were used to scan these 135 images at 34 locations. We divided these images into sets of pictures taken with the same scanner. Every image had a voxel size of $1 \times 1 \times 1 \text{ mm}^3$ and was strongly registered to MNI space. To create brain masks, we employed BET. Different patient

Table 1 Dataset description

Dataset	Source	Images	Voxel size mm^3
BT1	RSS	6	$0.49 \times 0.49 \times 0.80$
BT2	RSS	12	$1.25 \times 1 \times 1$
BT3	MRBrain challenge	5	$0.95 \times 0.95 \times 3$
BT4	IBSR	18	$0.84 \times 0.84 \times 1.5$
BT5	IBSR	20	$1 \times 3.1 \times 1$
WML1	RSS	20	$0.49 \times 0.49 \times 0.80$
WML2	MS Lesion challenge	20	$0.5 \times 0.5 \times 0.5$
WML3	MS Lesion challenge	20	$0.5 \times 0.5 \times 0.5$
HC1-27	ADNI	1-10	$1 \times 1 \times 1$
HC28-46	ADNI	1-10	$1 \times 1 \times 1$

groups constitute sources of domain shift when they differ in biological (age, sex, genetics, comorbidities), clinical (tumor subtype, stage, treatment), or technical (scanner/protocol) factors that change the underlying image distribution.

4.2 Data pre-processing

The data augmentation technique was used to adjust all images for intensity non-uniformity. The 4–96th percentile was then rescaled to the 0–1 interval within the brain mask to equalize all image intensities.

4.3 Feature analysis

The following characteristics were computed at all voxel levels and for every sequence: These variables include:

- (i) *Intensity*: it is computed after convolution with a squared exponential kernel with multiple σ_F ,
- (ii) Gradient magnitude of intensity following convolution with a Gaussian kernel function with different σ_F , and
- (iii) Intensity's laplacian following convolution with a Gaussian kernel function with different σ_F . The 4th–96th percentile scaling was chosen to stabilize intensity normalization across scans while avoiding outlier-driven distortions. Laplacian features are justified because they provide boundary-sensitive information that complements intensity-based features, and their inclusion is supported by both empirical results and prior neuroimaging literature.
- (iv) Spatial properties are exclusive to brain tissue. Due to the generally smaller voxel sizes, $\sigma_F = 1, 2$, and 4 mm^3 were selected for brain tissue classification, $\sigma_F = 0.5, 1$, and 2 mm^3 for WML, and $\sigma_F = 1, 2.2$, and 5 mm^3 for hippocampus segmentation which is comparable to the hippocampus segmentation approach. The spatial characteristics were introduced for brain tissue classification. The following formula was used to determine the cylindrical dimensions ($R, \theta_F, \text{and } z$) for every voxel: Initially, the source was placed as the center of each brain mask, and each mask ranged from -1 to 1 in each axis. In this case, z is the voxel's location in the cranial-caudal position, R is the space between the origin O and voxel, and the positive angle between the origin O and voxel is represented as θ_F (i.e. $\in [0, \pi]$) and the anterior–posterior axis. Since lesions may develop at different places in training and test data, spatial characteristics were not included for WML. Since spatial data is encrypted in the chosen ROI, no spatial characteristics were employed for hippocampal differentiation. Therefore, the brain tissue studies yielded 13 features, the WML investigations produced 30 features, and the

hippocampus investigations produced 10 features. Both image weighting and kernel learning took advantage of these attributes. Every feature was adjusted to zero mean, unit variance/image for every application. There was no feature decrease. Tissues (automatically trained kNN on multi-spectral MRI; Erasmus MC/RSS work): Similarity indices (Dice) 0.92–0.93 across tissues: CSF 0.93, GM 0.92, WM 0.92 (12 manually labeled cases; robustness shown on 59 subjects). Mean Dice for hippocampus ≈ 0.87 (cross-validated across sequences; same paper also reports cerebellum 0.95 but you asked hippocampus). Tissues (IBSR18): Dice ≈ 0.94 WM, 0.94 GM, 0.84 CSF reported by the NeuroNet paper evaluating on IBSR and Tissues (MRBrainS13, “BGR2” method on the official results page): Dice ≈ 0.884 WM, 0.847 GM, 0.783 CSF; also reported: whole brain 0.951, ICV 0.960. (Metrics are per-structure Dice on the public leaderboard.)

4.4 Experimental analysis

By training the entire images extracted from various datasets, the leave-one-out cross-validation was employed to classify the images of each dataset for each of the three applications. All of the voxels in an ROI within the hippocampus were categorized as one of the two kinds of hippocampus or non-hippocampus to perform hippocampus classification. Voxels inside the brain mask were categorized as WM, WML, GM, non-WML, and CSF, respectively, for brain tissue and WML classification. More information is provided about the voxel sampling in test and train samples. We employed nearly 60 base kernels for MKL, including the four kernel types are inverse square distance, Gaussian, Laplacian, and inverse distance were employed. We employed 15 distinct kernel parameter values like $\gamma = 10^{[-8, -7, \dots, 5, 6]}$ for each type of kernel. Based on the WML experiments and brain tissue, the classification error was used to gauge each classifier's effectiveness. Dice overlapping was used to quantify performance in the hippocampal experiments. A paired two-tailed t-test with an acceptable level of $P = 0.05$ was employed to evaluate the significance of variations between the two approaches in each of the three scenarios. Different classifiers for the left and right hippocampal regions were used in the hippocampus studies, and they performed better than a joint classification. Hippocampus classification efficiency was averaged on both sides of the hippocampal regions. The interior-reflective Newton approach was used to optimize the kernel and image weights. In the following optimization, image weights were less than 0.01, and kernel weights were set to null (i.e. 0). Furthermore, since classifier is more frequently employed in healthcare image classification than support vector machines, we conducted

trials using an DWAN classifier rather than an SVM. Based on earlier research on image weighting for healthcare image classification, we employed default settings for all other parameters. The MATLAB 2020a implementation technique was employed in this research. For the studies, we employed DWAN because existing methods are unable to process kernels explicitly. In this case, attributes for the training and test images were created by extracting the (d) main elements with the highest variance from kernel space. The total quantity of actual data features (WML (30 features), brain samples (13 features), and the hippocampus (10 features)) was the value for d .

4.5 Analysis with training and testing samples

Voxels inside the brain image were uniformly selected at random to create training and verification sets for brain cell classification. Only samples with a standardized image intensity greater than 0.750 were chosen for training and verification in the WML classification process because WMLs looked prominent on the samples. To preserve WML samples by discarding non-WML samples, a threshold of 0.75 was selected. Therefore, nearly all WM and WML samples were retained, but the majority of CSF and some GM voxels were eliminated. WML and non-WML samples were then randomly sampled into the training and test datasets, ensuring that 20% of the chosen voxels in each dataset were WML samples. This modification was made since classifiers will probably decide to categorize all samples as non-WML because of the extremely low probability for a sample to be WML. In contrast to the classification, a proportionally sampled dataset was used to derive the KL and image weights. All training voxels from several scanning devices were non-rigidly synchronized to the test image to segment the hippocampal region. The mapping of training images was converted appropriately to choose test samples, and an ROI made up of all the voxels that at least 10% of the atlases determined to be hippocampal was created. Similarly, training samples were chosen by constructing an ROI, of which the samples were employed as training samples, and non-rigidly registering all training images to one another. Employing the registration parameters, the registrations were carried out by optimizing standardized mutual data using the Elastix registration toolbox. One thousand randomly selected samples per image (training and test images) made up the training sets for image weighting and MKL for entire applications. To create the training sets for the classification algorithms, all weights ($w < 0.01$) were initially declared as zero. Then, based on the weight vector w , 10,000 voxels were chosen at random from all training samples. Every training image was given the same weight for the unweight classifications. The test set for hippocampal

segmentation included every voxel in the ROI. Ten thousand randomly chosen voxels for every test image made up the test sets for brain tissue and WML segmentation.

4.6 Weighted kernel learning

The parameters of the kernel and set for the methods in categories 0 and W (no MKL). The DWAN parameter is the only parameter that was required. Here, θ is the parameter of the MMD, θ and C were required for the MMD approaches. Adversarial learning and MMD-based kernel learning both enforce domain alignment but via different mechanisms. Integration is done by adding an MMD penalty into the adversarial training objective, where MMD provides kernel-based stability and adversarial loss provides adaptive flexibility. The result is more stable and effective distribution matching in segmentation and classification tasks. On every training image, these parameters were modified using leave-one-dataset-out grid searching. Therefore, a classification system trained on all other training samples from other datasets was used to identify the samples of each training image to determine the appropriate parameters for each test sample. Each of the three parameters was adjusted without image weighting for computing reasons. A weighted method greatly exceeded the baseline approach in all three applications. The most effective weighting technique varied depending on the application. The kernel leaning weighting significantly performed better than other weighting techniques for segmentation. The weighting approach was based on general image similarity instead of similarity which was probably caused by the extremely small number of samples. The samples are situated in an area with low $P(x)$ because they exhibit extreme magnitude in the scan, but other samples do not.

The kernel learning weighting approach works better than BD and MMD techniques for the weighting process because it concentrates more on regions of the variance with low values of $P(x)$, which is due to the log computation (variations in the training and verification distributions in areas where P_{m_i} is minimal contribute more to the kernel learning than in positions where the value of P_{m_i} is huge). KL and BD performed most effectively in brain cell segmentation and hippocampus differentiation, respectively, while the other two weighting approaches had less variation in these categories. MMD placed in the second position in both instances. When applied alone, image weighting performed better than kernel learning, particularly for brain tissue segmentation. Except for hippocampus differentiation, where efficiency was comparable among the image weighting combined with KL and exclusive of KL, integrating the two methods resulted in a little enhancement of overall efficiency. When evaluated with an independent KL

and MMD image weighting approach, where was no apparent improvement in performance when kernels and images were weighted jointly. Although the weights for classification were slightly distinct. The figure demonstrate that both approaches used the same base kernels for brain cells and WML classification. The assigned significant weights to kernels low-parameters (i.e., concentrates on variations across all observations) for the hippocampal studies. However only around 1/3 of the categorized images exhibited this behavior, and it did not seem to have a big impact on performance. The image weight distribution when MKL and image weighting were combined; for all three applications, they were quite similar to those computed without MKL.

To make comparisons easier, the additional material provides the effectiveness of the two classifications. Generally, DWAN benefited more from image weighting than existing approaches. KL image weighting typically outperformed the other classifiers, improving tissue classification for other approaches significantly and WML segmentation

non-significantly. No improvement in hippocampal performance was observed, which is not noteworthy because the variations among the various approaches were likewise negligible for this use case. The kernel learning performance for tissue and hippocampal differentiation was significantly inferior to SVM's. This is most likely because DWAN operates explicitly in kernel space, whereas kernel was used to gather features. The difference in the kernel learning spaces is 1 in all directions, however, it is ideal to collect adequate attributes to explain approximately all of the variation (e.g., 99%). Therefore, based on the application and dataset, approximately one thousand features were required to explain 99% of the variability. With so many features, the existing approaches did not perform well. Because of the same impact, the integration of kernel learning with image weighting provides less performance compared to other approaches for tissue identification Figs. 2, 3, 4 and 5.

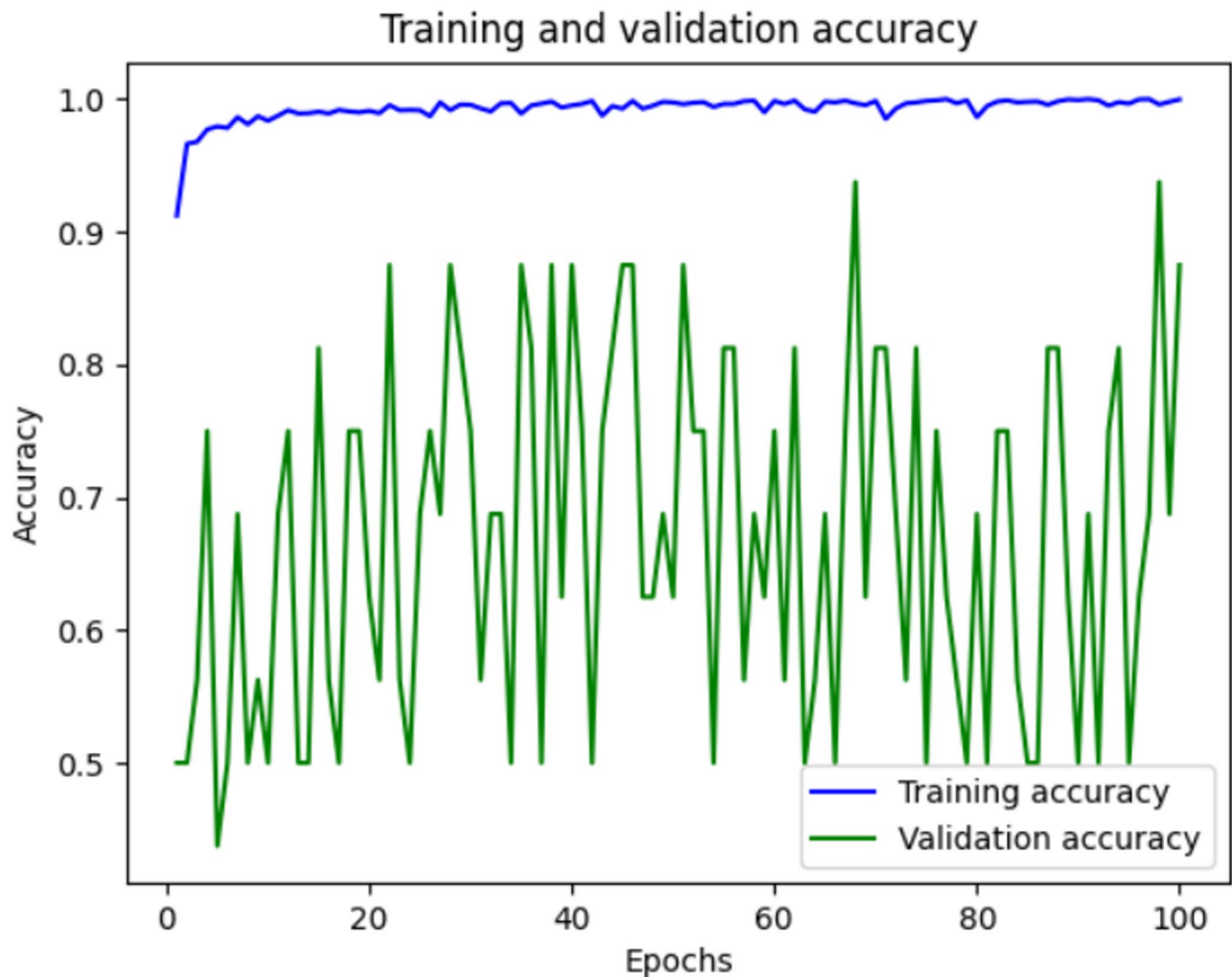
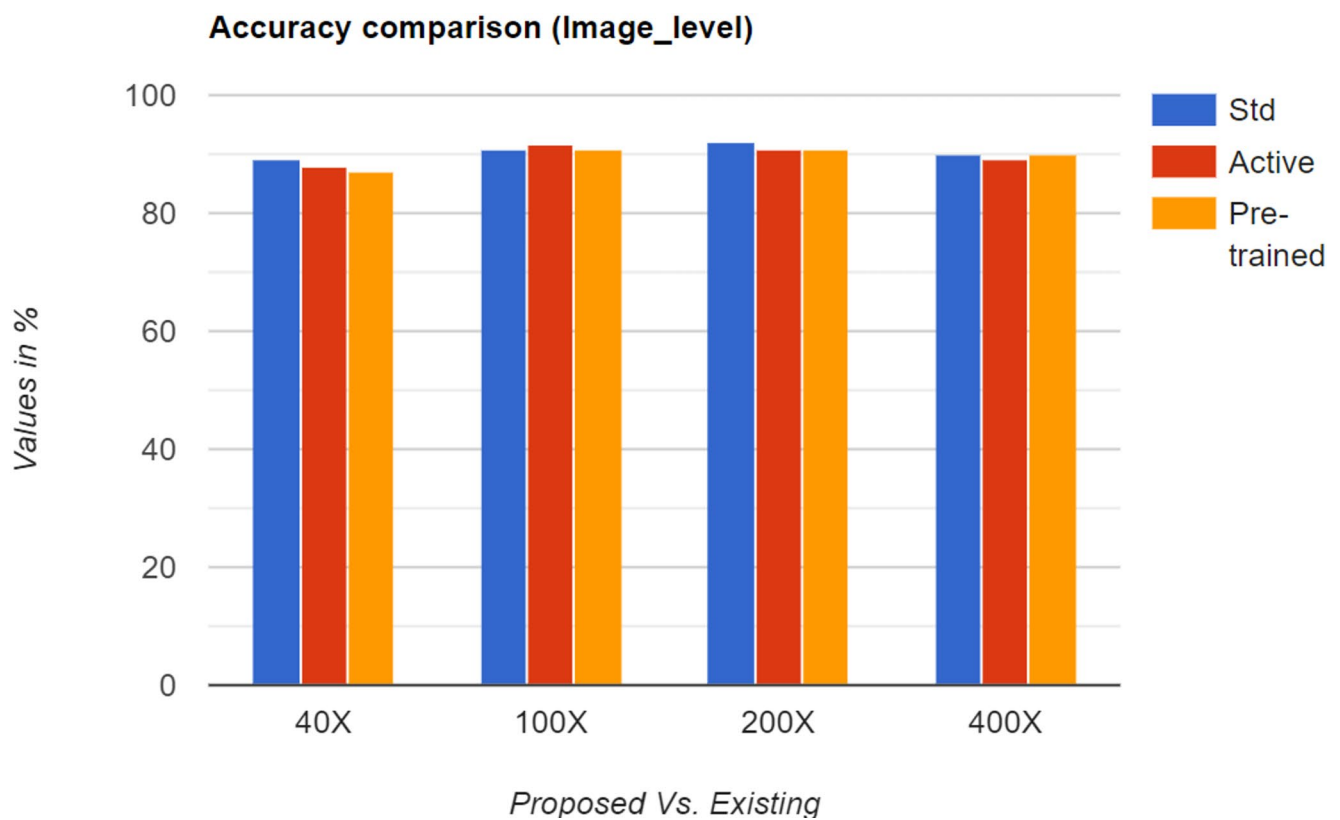
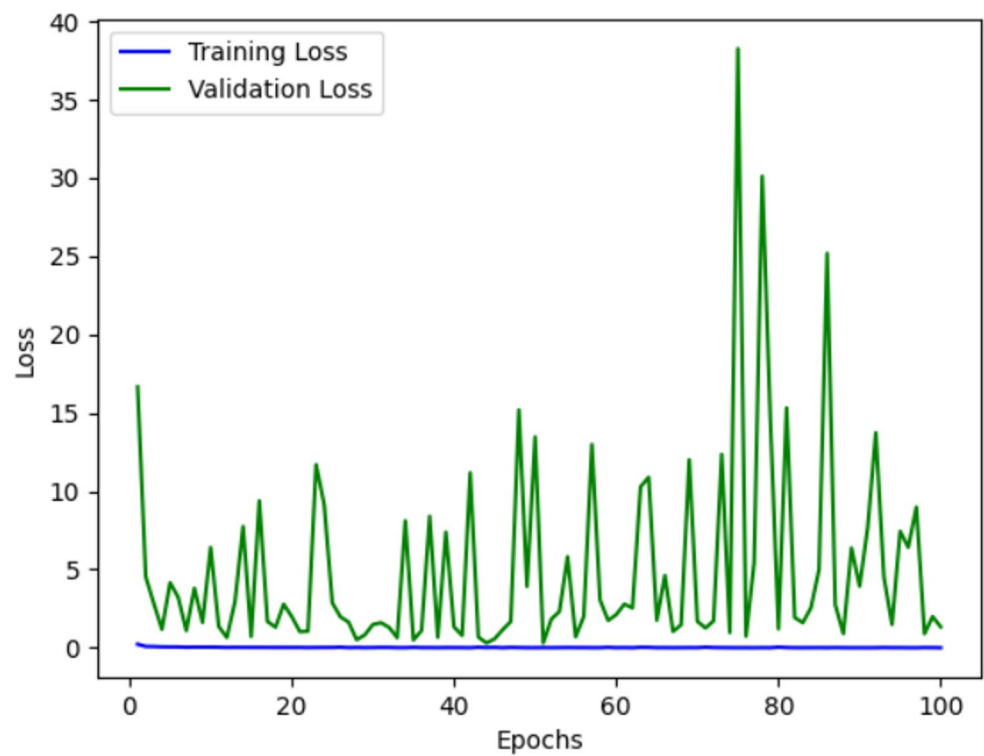


Fig. 2 Training and validation accuracy

Fig. 3 Training and validation loss**Fig. 4** Accuracy comparison (image level)

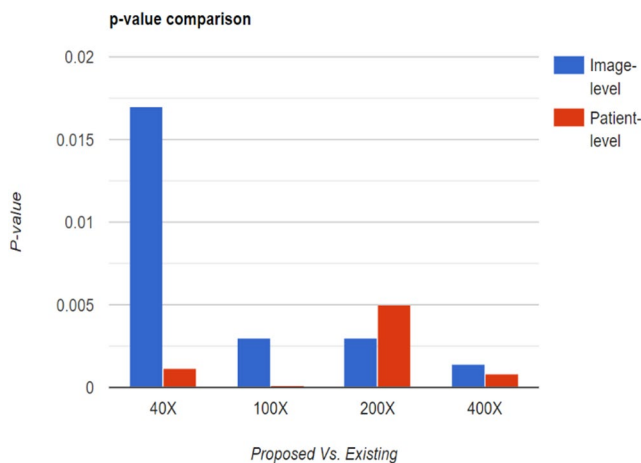


Fig. 5 *p*-value comparison

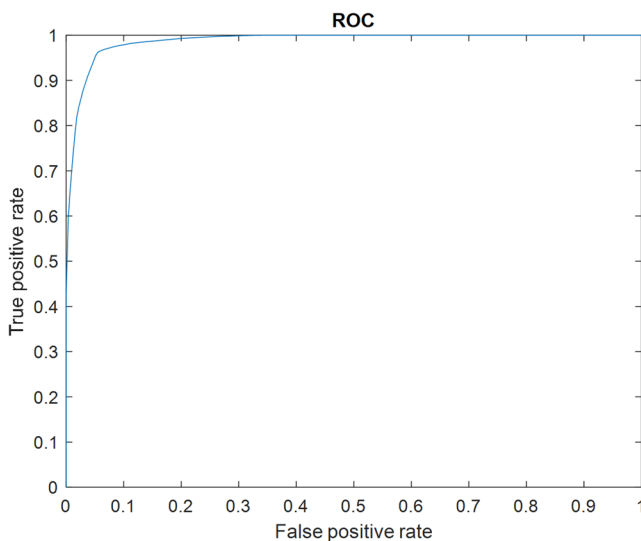


Fig. 6 ROC comparison

4.7 Computational requirements

Every test was run as a single-core task on a Windows OS (2.3 gigahertz) CPU. The KL techniques needed roughly 6 GB of storage capacity, while the non-kernel learning approaches needed 2–3 GB of memory. The mean calculation period for DWAN identification of a single test sample for each of the programs and techniques is provided. For the DWAN classifier, the baseline approach required an average of 8–14 s to categorize a single test image. According to the application and the source data, the various weighting techniques varied in duration from 16 s to slightly more than 4 min; bigger training datasets need longer computation times. Since gradient descent is used in each of these techniques, the computation time is based on how hard it is to obtain the best answer. The overall fastest method was followed by MMD and KL. The computation time increased

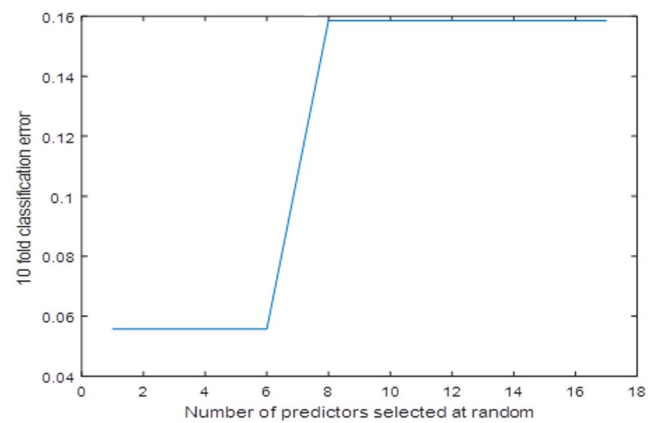


Fig. 7 Classification error

significantly for all methods that include kernel learning. In this case, the amount of training photos used quadratically affects the calculation time. The kernel learning process for the WML trials (20–30 training samples) takes roughly forty-five minutes. It took around two hours and forty minutes to classify the brain tissue (41–56 training voxels). This requires roughly 73 h and 10 min for the classification of the hippocampal region (128–134 training images). Using only a few fundamental kernels would significantly speed up these approaches, which are costly due to the computation of the 60 primary kernels for a single training sample. Depending on how many features were retrieved using data augmentation to acquire 30, 10, and 13 features it takes approximately 10–30 s. Figures 2–7 shows the experimental outcomes.

For the three applications under study are voxel-wise brain cells, WML, and hippocampus classification are image weighting considerably outperformed weighting all training images equally. This compelling advantage of image weighting is consistent with earlier research. KL divergence weighting outperformed other weighting techniques for WML classification, which had extremely unstable classes. Given that it assigns greater weight to portions of the distribution with a tiny prior distribution $P(x)$, KL weighting most likely performed better in this case than other weighting techniques. The best techniques varied for segmenting brain tissue and the hippocampus, and the variations among the three weighting techniques were not very significant. In general, the outcomes we achieved on several tasks demonstrate the benefits of integrating image weighting and kernel learning for a cross-scanner segmentation. In summary, the integration of feature representation transfer via image weighting and kernel learning seems to be a hopeful approach for supervised classification with heterogeneous training features that differ from the test samples. The positive outcomes of several techniques suggest that

the suggested techniques can be used for a variety of health-care image classification applications.

5 Conclusion

Thus, we analyze the supplementary utility of kernel learning for image weighting and look into how it may be used to lessen the discrepancies between data used for training and testing. Additionally, we provide a novel Dynamic Weighted Adversarial Network (DWAN) approach to image weighting that reduces the maximum mean difference (MMD) among test and training data, allowing for the mutual optimization of kernel and image weights. Employing both image weighting and kernel learning independently significantly improves performance on heterogeneous data, according to observations on brain cells and hippocampal differentiation. In this case, MMD weighting performs comparably to earlier suggested image weighting technique known as Dynamic Weighted Adversarial Network (DWAN). Performance can be slightly increased by combining kernel learning with image weighting, which can be adjusted separately or in conjunction. However, in future, hybrid optimization with learning approaches will be proposed for reducing the computational complexity.

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Declarations

Conflicts of interest As the corresponding author of this manuscript, the authors and Co-authors declare that they have no conflicts of interest.

Informed consent In this article, there is no human or animal sample was involved in this study.

Human and animals rights This article does not contain any studies involving animals performed and any studies involving human participants performed by any of the authors.

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